

Ahmed Amr

Machine Learning & Computer Vision Engineer

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[LinkedIn](#) • [GitHub](#) • [Portfolio](#)

Profile

I build end-to-end intelligent solutions that unlock the true potential of data. Specializing in deep Exploratory Data Analysis (EDA), robust Machine Learning architectures, and Computer Vision, I manage the full AI lifecycle—from processing complex raw data to deploying scalable, production-ready models

Education

Bachelor of Computer Science at Modern Academy Maadi, Cairo, Egypt

September 2022 - Present

Internships

Data Science & Machine Learning Intern at [IT-Gate Academy](#), Cairo, Egypt

July 2024 - August 2024

Completed intensive training in Data Science fundamentals and practical machine learning. Gained hands-on experience in exploratory data analysis, data preprocessing, feature engineering, and building & evaluating ML models on real-world dataset

Skills

Programming & Databases: Python, SQL

Data Science & Analysis: NumPy, Pandas, SciPy, Scikit-learn, Matplotlib

AI & Computer Vision: PyTorch, OpenCV, YOLO, U-Net, Image Segmentation, Object Detection

Backend & Tools: Flask, Docker, Git, GitHub

Projects

Axon: End-to-End Alzheimer's Detection System • [GitHub](#) • [Live Demo](#)

January 2026 - June 2026

Designed and developed an end-to-end medical AI solution classifying brain MRI scans into 4 stages of Alzheimer's Disease with 93.2% accuracy. Engineered a custom CNN architecture using TensorFlow/Keras, trained on 33,984 augmented MRI images with BatchNorm, Dropout, and MobileNetV2 normalisation. Achieved 99% precision/recall on Moderate Demented cases, critical for urgent medical intervention. Built a Flask REST API for real-time inference and integrated it with a Flutter mobile application. Deployed the model to Hugging Face Spaces using Gradio for public accessibility

Customer Churn Prediction • [GitHub](#)

May 2026

Built 6 ML models (both built-in and from scratch) to predict customer churn on 440K records. Conducted deep EDA and feature engineering. Best model: Random Forest at 93% accuracy.

Food Image Classifier • [GitHub](#) • [Live Demo](#)

May 2026

Fine-tuned ResNet50 on the Food-101 dataset (101K images, 101 classes) for food classification and calorie estimation. Deployed as an interactive web app on Hugging Face Spaces.

Certificates

Statistics 101 from Cognitive Class • [Link](#)

November 2025

Data Analysis with Excel Pivot Tables from 365 Data Science • [Link](#)

November 2025

Scientific Computing with Python from freeCodeCamp • [Link](#)

March 2026